CampusFixx — Project Overview & Features

CampusFixx is a real-time, digitized campus complaint and maintenance management platform .It replaces slow, manual campus maintenance workflows with an automated, transparent, multi-role lifecycle system built on **Firebase Fire store** and **EmailJS**.

1. Multi-Role Portals & Access Control

A. **Student Portal**

- **Verified Student Authentication :** Validates student Name, Register Number, and Department against the Firestore students collection before granting access.
- **Smart Issue Reporting :**
  - Auto-fills student name, registration number, and department.
  - Captures detailed location: Block (*Additional Block, Main Block, Temple Tower, PIF*), Floor, Room/Lab number, and nearest Landmark.
  - Category picker (Electrical, Plumbing, Cleanliness, Internet, Furniture, Other) with custom free-text input if "Other" is selected.
  - Priority assignment: *Low*, *Medium*, *High (Urgent)*.
- **Daily Fair Usage Policy / Anti-Spam :** Restricts complaint submissions to a maximum of **5 complaints per day per student**.
- **Personal Complaint History :**
  - Dedicated dashboard displaying all past and current tickets submitted by the logged-in student.
  - Real-time status badges (*Pending*, *Resolved*, *Denied*, *Re-opened*, *Verified*).
- **Two-Way Resolution Verification :**
  - When an admin marks an issue as "Resolved", the student receives verification controls:
    - **"Problem Fixed"** → Moves ticket to **Verified**.
    - **"Not Fixed"** → Reopens the ticket as **Re-opened** for immediate re-escalation.

B. **Staff & Faculty Portal**

- **Dedicated Faculty Login :** Independent authentication route for teachers and institutional staff.
- **Faculty Complaint Submission :** Tailored interface allowing staff to report classroom, lab, departmental, or administrative facility breakdowns without student registration constraints.

C. **Department In-Charge Portal**

- **Role-Based In-Charge Login :** Dedicated logins for departmental heads (*Electrical, Plumbing, Cleanliness*).
- **Scoped Department View :** In-charges only see and manage complaints categorized under their specific department, keeping the interface focused and preventing cross-department confusion.

D. **Super Admin Portal**

- **Master Admin Login :** Full administrative access across all campus departments.
- **Real-time KPI Dashboard :**
    - Live stats counters for **Total**, **Pending & Re-opened**, **Resolved**, and **Denied** tickets.
- **Multi-Filter & Tab System :**
    - Category tabs: *All Issues, Electrical, Plumbing, Cleanliness, Internet, Furniture, Others*.
    - Status filter (*Pending, Resolved, Denied*) and Priority filter (*High, Medium, Low*).
- **Live Global Search :** Instant search across Complaint ID, Student Name, and Issue Description.
- **Status Action Controls :**
    - One-click **Resolve** or **Deny**.
    - Rejection prompt capturing an admin explanation saved as admin Feedback, visible to the reporter.
- **Audit Trail / History:** Displays timestamps for ticket creation, updates, and resolution dates directly on ticket cards.

2. Intelligent & Automated Capabilities

- **AI-Powered Keyword Categorization :**
    - Analyzes issue titles and descriptions using NLP keyword detection (e.g., detecting *bulb, wire, trip, light* → Electrical; *leak, tap, pipe, toilet* → Plumbing; *wifi, lan, signal* → Internet; *bench, chair, desk* → Furniture).
    - Automatically assigns the right department tag even if the user selected a generic category.

- **Automated Email Routing via EmailJS :**
    - Automatically sends an email notification with full incident details (student info, exact location, description) directly to the specific department head upon submission.
- **SLA Tracking & Auto-Escalation Engine (admin.js):**
    - Tracks ticket age against configurable Service Level Agreements (SLAs) based on priority:
        - *High Priority:* 1 minute (hackathon demo scale)
        - *Medium Priority:* 2 minutes
        - *Low Priority:* 3 minutes
    - Overdue tickets are automatically tagged with escalated: true, visually flagged with ⚠️ **ESCALATED** badges, and trigger alert toast notifications on the dashboard.
- **CSV Export & Audit Reporting :**
    - Generates and downloads a clean, dated CSV spreadsheet containing all complaints, locations, statuses, and student details for institutional reviews.

3. **Direct Tracking & Administrative Utilities**

- **Standalone Complaint Status Tracker :**
    - Quick lookup by Unique Complaint ID (CMP-YYYYMMDD-XXXXX) without needing to log in.
    - Shows real-time status, location, admin remarks, and verification buttons.
- **Bulk Student Importer :**
    - Client-side CSV batch parser powered by PapaParse.
    - Enables administrators to bulk-upload thousands of student records into Firebase Firestore with live progress logging.

4. Design & Tech Stack

Component	Technology / Technique
Frontend	Vanilla HTML5, Modern CSS3, JavaScript (ES6+)
Styling & Theme	Deep Blue Glassmorphism (style.css), Plus Jakarta Sans typography, custom CSS animations, mobile-responsive layout
Database	Google Firebase Firestore (Real-time onSnapshot listeners)
Notification Services	EmailJS Browser SDK
Data Processing	PapaParse (CSV Parsing & Bulk Ingestion)

FUTURE ENHANCEMENT

1. Advanced AI & Computer Vision

- **Image-Based Reporting**: Enable students to upload photos of the issue (e.g., a leaking pipe or broken bench). Use **Google Cloud Vision AI** to automatically verify the damage and its severity from the image.
- **Gemini AI Integration**: Replace keyword-based routing with a Large Language Model (LLM) to handle complex descriptions and even predict the specific tools a technician needs to bring before they arrive.

2. Physical Infrastructure & IoT

- **Smart Sensors (IoT)**: Integrate IoT sensors in water tanks and electrical rooms to **automatically raise a complaint** the moment a leak or power surge is detected—fixing issues before students even notice them.
- **QR Code Room ID**: Place unique QR codes in every room/lab. Students can just scan the code, and the room number and block will be auto-filled in the form instantly.

3. Service Optimization

- **Maintenance Staff App**: A dedicated mobile application for technicians with **Push Notifications** and GPS navigation to the exact room location.
- **Predictive Maintenance**: Use historical data to predict which blocks are likely to have issues (e.g., "Block A usually has plumbing issues every 3 months") and schedule preventive check-ups.

4. Advanced User Experience

- **In-App Real-Time Chat**: Allow a direct chat window between the student and the assigned technician for quick clarifications or to share location details.
- **Gamification/Rewards**: Implement a "Maintenance Leaderboard" where departments get awards or higher budgets for maintaining a "Zero Escalation" streak.

5. Management Analytics

- **Heatmap Visualization**: An admin view showing a color-coded map of the campus. Red zones indicate high complaint density, helping management identify where infrastructure needs a complete overhaul.
- **Multilingual Support**: Support for local languages to ensure that all staff members, including non-teaching staff, can report and resolve issues comfortably.